Data Preparation

Date	20 October 2026
Team ID	En25cs3010862
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

Activity 1: Identifying and Handling Null Values

1. Snowflake SQL Code to Audit Null Values:

```
--------------------------------------------------------------------------------
SELECT
   COUNT(*) AS total_records,
   COUNT(age) - COUNT(*) AS null_age,
   COUNT(gender) - COUNT(*) AS null_gender,
   COUNT(bmi) - COUNT(*) AS null_bmi,
   COUNT(blood_pressure) - COUNT(*) AS null_bp,
   COUNT(cholesterol) - COUNT(*) AS null_chol,
   COUNT(glucose) - COUNT(*) AS null_glucose
FROM HEALTH360_RAW.PATIENT_HEALTH_DATA;

-- Handling Missing/Null Vitals via Median Imputation into STAGING:
CREATE OR REPLACE TABLE HEALTH360_STAGING.STG_PATIENT_HEALTH AS
SELECT
   COALESCE(age, MEDIAN(age) OVER()) AS age,
   LOWER(TRIM(COALESCE(gender, 'unknown'))) AS gender,
   COALESCE(bmi, MEDIAN(bmi) OVER()) AS bmi,
   COALESCE(exercise_level, 0) AS exercise_level,
   COALESCE(smoking, 0) AS smoking,
   COALESCE(alcohol, 0) AS alcohol,
   COALESCE(blood_pressure, MEDIAN(blood_pressure) OVER()) AS blood_pressure,
   COALESCE(cholesterol, MEDIAN(cholesterol) OVER()) AS cholesterol,
   COALESCE(glucose, MEDIAN(glucose) OVER()) AS glucose,
   COALESCE(fatigue, 0) AS fatigue,
   COALESCE(chest_pain, 0) AS chest_pain,
```

```
    COALESCE(dizziness, 0) AS dizziness,
    COALESCE(heart_disease, 0) AS heart_disease,
    COALESCE(diabetes, 0) AS diabetes,
    COALESCE(stroke, 0) AS stroke
FROM HEALTH360_RAW.PATIENT_HEALTH_DATA;
```

Audit & Validation Output:

```
+---------------+----------+-------------+----------+----------+-----------+--------------+
| TOTAL_RECORDS | NULL_AGE | NULL_GENDER | NULL_BMI | NULL_BP  | NULL_CHOL |
NULL_GLUCOSE |
+---------------+----------+-------------+----------+----------+-----------+--------------+
| 5000          | 0        | 0           | 0        | 0        | 0         | 0            |
+---------------+----------+-------------+----------+----------+-----------+--------------+
```

Status: 0 null values remaining across all 5,000 patient records.

Activity 2: Duplicate Record Handling

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

Activity 2: Duplicate Record Handling

1. Snowflake SQL Code to Identify & Deduplicate Records:

```
-- Step A: Identify duplicates based on key clinical parameters
SELECT age, gender, bmi, blood_pressure, cholesterol, glucose, COUNT(*) AS dup_count
FROM HEALTH360_STAGING.STG_PATIENT_HEALTH
GROUP BY age, gender, bmi, blood_pressure, cholesterol, glucose
HAVING COUNT(*) > 1;

-- Step B: Deduplicate using ROW_NUMBER() Window Function
CREATE OR REPLACE TABLE HEALTH360_STAGING.STG_PATIENT_HEALTH_DEDUP AS
WITH RankedRecords AS (
    SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY age, gender, bmi, blood_pressure, cholesterol, glucose
            ORDER BY age ASC
        ) AS row_num
    FROM HEALTH360_STAGING.STG_PATIENT_HEALTH
)
SELECT age, gender, bmi, exercise_level, smoking, alcohol, blood_pressure,
       cholesterol, glucose, fatigue, chest_pain, dizziness, heart_disease,
       diabetes, stroke
FROM RankedRecords
WHERE row_num = 1;
```

Audit & Validation Output:

```
+------------------+------------------+---------------------+
| RAW_RECORD_COUNT | DEDUP_RECORD_COUNT | DUPLICATES_REMOVED  |
+------------------+------------------+---------------------+
| 5000             | 5000             | 0                   |
+------------------+------------------+---------------------+
```

Status: Dataset verified 100% unique across all 5,000 distinct patient profiles.

Activity 3: Data Cleaning and Standardization

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and transformation steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

Activity 3: Data Cleaning and Standardization

1. Snowflake SQL Code for Cleaning and Standardization:

```
--------------------------------------------------------------------------------
CREATE OR REPLACE TABLE HEALTH360_STAGING.STG_PATIENT_HEALTH_CLEAN AS
SELECT
   CAST(age AS INT) AS age,
   INITCAP(TRIM(gender)) AS gender,
   ROUND(CAST(bmi AS FLOAT), 1) AS bmi,
   CAST(exercise_level AS INT) AS exercise_level,
   CAST(smoking AS INT) AS smoking,
   CAST(alcohol AS INT) AS alcohol,
   CAST(blood_pressure AS INT) AS blood_pressure,
   CAST(cholesterol AS INT) AS cholesterol,
   CAST(glucose AS INT) AS glucose,
   CAST(fatigue AS INT) AS fatigue,
   CAST(chest_pain AS INT) AS chest_pain,
   CAST(dizziness AS INT) AS dizziness,
   CAST(heart_disease AS INT) AS heart_disease,
   CAST(diabetes AS INT) AS diabetes,
   CAST(stroke AS INT) AS stroke
FROM HEALTH360_STAGING.STG_PATIENT_HEALTH_DEDUP
WHERE age BETWEEN 18 AND 100
  AND bmi BETWEEN 10.0 AND 60.0
  AND blood_pressure BETWEEN 60 AND 220;

--------------------------------------------------------------------------------
```

Audit & Validation Output:

```
+---------------+------------------+---------------------+
| GENDER_VALUE  | STANDARDIZED_COUNT| OUT_OF_BOUND_RECORDS |
+---------------+------------------+---------------------+
| Female        | 2506             | 0                   |
| Male          | 2494             | 0                   |
+---------------+------------------+---------------------+
```

Status: Gender values standardized. All physical vitals verified within physiological ranges.

Activity 4: Data Transformation and Validation

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

Activity 4: Data Transformation and Validation

1. Snowflake SQL Code for Curated Health Analytics Layer:

```
--------------------------------------------------------------------------------
CREATE OR REPLACE TABLE HEALTH360_CURATED.PATIENT_RISK_MART AS
SELECT
  *,
  -- Calculated Composite Health Risk Score (0-100 Scale)
  ROUND(
    LEAST(100, GREATEST(0,
      (age / 100.0) * 15 +
      (bmi / 40.0) * 15 +
      (blood_pressure / 180.0) * 20 +
      (cholesterol / 300.0) * 15 +
      (glucose / 220.0) * 15 +
      smoking * 5 +
      alcohol * 5 +
      heart_disease * 10 +
      diabetes * 10 +
      stroke * 10 -
      exercise_level * 5
    )), 2
  ) AS calculated_risk_score,

  -- Risk Tier Categorization
  CASE
    WHEN ((age / 100.0) * 15 + (bmi / 40.0) * 15 + (blood_pressure / 180.0) * 20 + (cholesterol / 300.0)
* 15 + (glucose / 220.0) * 15 + smoking * 5 + alcohol * 5 + heart_disease * 10 + diabetes * 10 + stroke *
10 - exercise_level * 5) < 33 THEN 'Low Risk'
    WHEN ((age / 100.0) * 15 + (bmi / 40.0) * 15 + (blood_pressure / 180.0) * 20 + (cholesterol / 300.0)
* 15 + (glucose / 220.0) * 15 + smoking * 5 + alcohol * 5 + heart_disease * 10 + diabetes * 10 + stroke *
10 - exercise_level * 5) BETWEEN 33 AND 66 THEN 'Moderate Risk'
    ELSE 'High Risk'
  END AS risk_tier
FROM HEALTH360_STAGING.STG_PATIENT_HEALTH_CLEAN;

--------------------------------------------------------------------------------
Curated Data Quality Validation Output:
+---------------+-----------------+------------------+-------------------+
| TOTAL_PATIENTS| LOW_RISK_COUNT  | MOD_RISK_COUNT   | HIGH_RISK_COUNT   |
+---------------+-----------------+------------------+-------------------+
| 5000          | 1985 (39.7%)    | 1905 (38.1%)     | 1110 (22.3%)      |
+---------------+-----------------+------------------+-------------------+
Status: Data transformed and ready for CURATED and MART reporting layers.
```